FIITJEE NSEC MOCK TEST-4

Time Allotted: 2 Hours | **CODE:** | **Maximum Marks: 216**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

Write the question paper code (mentioned above) on YOUR OMR Answer Sheet (in the space provided), otherwise your Answer Sheet will NOT be evaluated. Note that the same Question Paper Code appears on each page of the question paper.

Instructions to Candidates:

1. Use of mobile phone, smart watch, and ipad during examination is STRICTLYPROHIBITED.
2. In addition to this question paper, you are given OMR Answer Sheet along with Candidate's copy.
3. On the OMR sheet, make all the entries carefully in the space provided **ONLY** in **BLOCK CAPITALS** as well as by properly darkening the appropriate bubbles.

   Incomplete/ incorrect/ carelessly filled information may disqualify your candidature.
4. On the OMR Answer Sheet, use only **BLUE** or **BLACK BALLPOINT PEN** for making entries and filling the bubbles.
5. **Your Enrollment number entered** on the OMR Answer Sheet shall remain your login credentials means login id and password respectively for accessing your performance / result in NSE.
6. Question paper has two parts. In part A1 (Q. No.1 to 48) each question has four alternatives, out of which only one is correct. Choose the correct alternative (s) and fill the appropriate bubble (s), as shown.

Q. No. 22	a	⬬	c	d

   In part A2 (Q. No. 49 to 60) each question has four alternatives out of which any number of alternative (s) (1, 2, 3, or 4) may be correct. You have to choose all correct alternative(s) and fill the appropriate bubble(s), as shown

Q. No. 54	a	⬬	c	⬬
7. For **Part A1**, each correct answer carries 3 marks whereas 1 mark will be deducted for each wrong answer. In **Part A2**, you get 6 marks if all the correct alternatives are marked. No negative marks in this part.
8. Rough work may be done in the space provided. There are 15 printed pages in this paper
9. Use of **non - programmable scientific** calculator is allowed.
10. No candidate should leave the examination hall before the completion of the examination.
11. After submitting answer paper, take away the question paper & Candidate's copy OMR sheet for your reference.

Please DO NOT make any mark other than filling the appropriate bubbles properly in the space provided on the OMR answer sheet.

OMR answer sheets are evaluated using machine, hence CHANGE OF ENTRY IS NOT ALLOWED. Scratching or overwriting may result in a wrong score.

DO NOT WRITE ON THE BACK SIDE OF THE OMR ANSWER SHEET.

Name of the Candidate :____

Batch :____________________ **Date of Examination :**____________

Enrolment Number :____

INDIAN ASSOCIATION OF PHYSICS TEACHERS
NATIONAL STANDARD EXAMINATION IN CHEMISTRY
(NSEC-2023)

Time: 120 minutes **Max. Marks: 216**

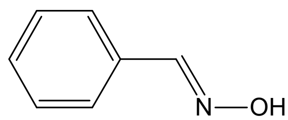
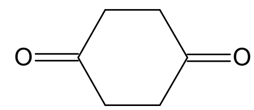
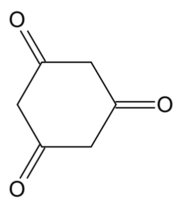
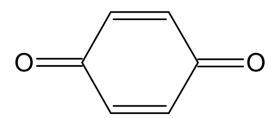
Attempt All Sixty Questions

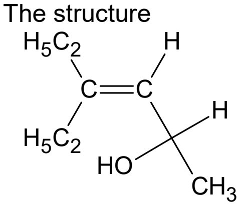
A – 1

ONLY ONE OUT OF FOUR OPTIONS IS CORRECT BUBBLE THE CORRECTION OPTION

1. Ionization of hydrogen atom gives
(A) hydride ion (B) hydronium ion
(C) proton (D) hydroxyl ion

2. The element whose electronic configuration is $1s^2$, $2s^2$, $2p^6$, $3s^2$ is a/an
(A) metal (B) inert gas
(C) metalloid (D) non-metal

3. Tautomerism is not exhibited by
(A) 
(B) 
(C) 
(D) 

4. The structure

Shows
(A) geometrical isomerism (B) optical isomerism
(C) tautomerism (D) both geometrical & optical isomers

5. de Broglie wavelength of electrons emitted by a metal whose threshold frequency is 2.25×10^{14} Hz when exposed to visible radiation of wavelength 500 nm is of the order is
(A) 1 Å (B) 1 nm
(C) 1 pm (D) 1 fm

Space for rough work

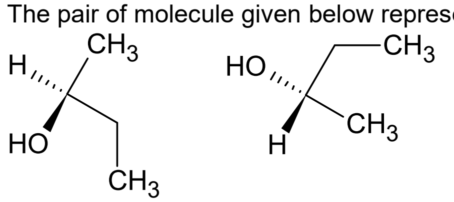
6. An electron will have the highest energy in the set

(A) $3, 2, 0, +\frac{1}{2}$
(B) $4, 0, 0, +\frac{1}{2}$
(C) $3, 0, 0, +\frac{1}{2}$
(D) $4, 0, 0, -\frac{1}{2}$

7. Addition of HCl to propene involves

(A) $CH_3\dot{C}H_2CH_2Cl\,CH$
(B) $CH_3-CH_2-\overset{+}{C}H_2$
(C) $CH_3-\ddot{C}HCH_2Cl$
(D) $CH_3-\overset{+}{C}H-CH_3$

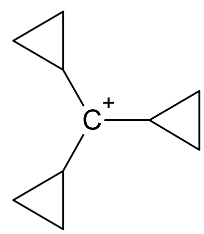
8. The pair of molecule given below represents



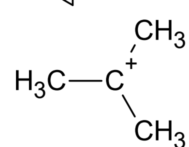
(A) Enantiomers
(B) Diastereomers
(C) Regiomers
(D) None of these

9. Which of the following carbocation is most stable?

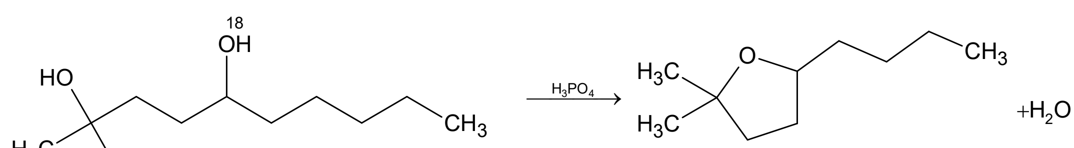
(A) $Ph-\overset{+}{C}H_2$

(B) 

(C) $Ph-\overset{+}{C}H-Ph$

(D) 

10.

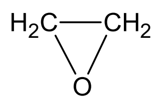


(A)

^{18}O will be in

(A) (A)
(B) H_2O
(C) both
(D) none

Space for rough work

11.  $\xrightarrow[\text{(ii) } H_2O]{\text{(i) } CH_3MgCl}$ X, Here X is –

(A) CH_3CH_2OH
(B) $(CH_3)_2CHOH$
(C) $CH_3CH_2CH_2OH$
(D) $HO-CH_2-CH_2-CH_2-CH_2-OH$

12. In S_{N_1} reaction, racemisation occurs because of
(A) Carbocation formation
(B) Carbanion formation
(C) Enolisationic
(D) None of these

13. The equilibrium constant for the reaction
$$2SO_{2(g)} + O_{2(g)} \rightleftharpoons 2SO_{3(g)}$$
is 5. If the equilibrium mixture contains equal moles of SO_3 and SO_2, the equilibrium partial pressure of O_2 gas is:
(A) 0.2 atm
(B) 2 atm
(C) 0.02 atm
(D) 0.04 atm

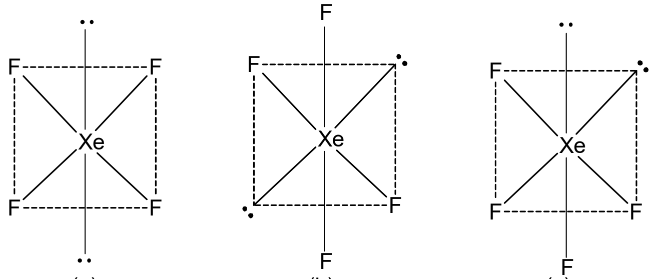
14. At a certain temperature the $\left[OH^-\right]$ in water is 10^{-8} mole/litre now some NaOH is added in water so that the $\left[OH^-\right]$ becomes 10^{-6} mole/litre then the $\left[H^+\right]$ in new solution.
(A) 10^{-11}
(B) 10^{-7}
(C) 10^{-10}
(D) 10^{-14}

15. pH of 0.1 M monoacidic base solution is found to be 12 Thus its osmotic pressure at TK is
(A) 0.01 RT
(B) 0.11 RT
(C) 0.001 RT
(D) 1.11 RT

16. Which of the following has zero dipole moment?
(A) BF_3
(B) BeClBr
(C) CH_2Cl_2
(D) COS

Space for rough work

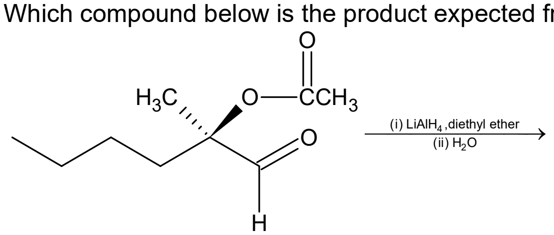
17. The XeF_4 structure can be represented



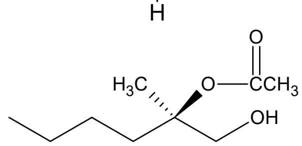
(a) (b) (c)

(A) by (a) only
(B) by (a) and (b) both
(C) by (a), (b) and (c)
(D) by (a) and (c) only

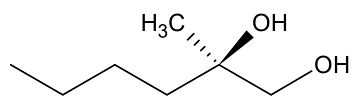
18. Which compound below is the product expected from the following reaction?



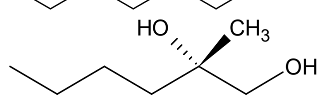
(A)



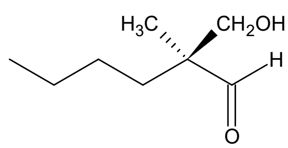
(B)



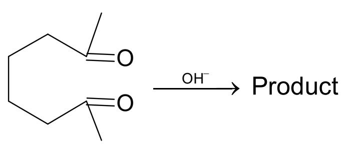
(C)



(D)

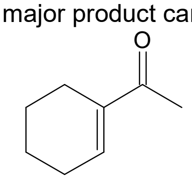


19.



The major product can be

(A)



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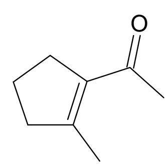

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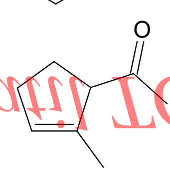
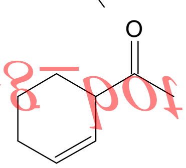
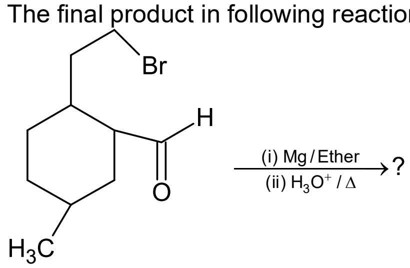


(D)

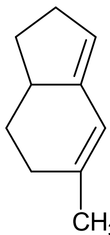


Space for rough work

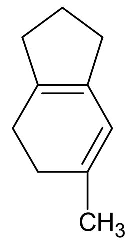
20. The final product in following reaction is



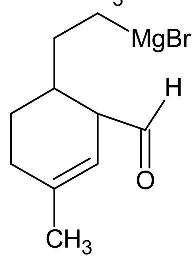
(A)



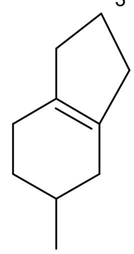
(B)



(C)

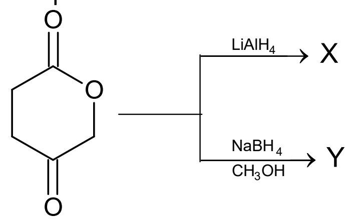


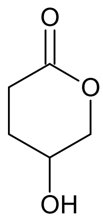
(D)

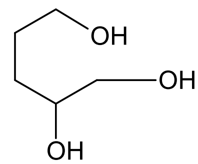


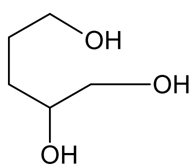
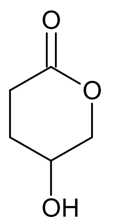
Space for rough work

21. The products X and Y in the reactions given below are



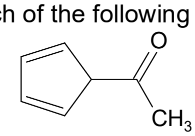
(A)  (Both X and Y)

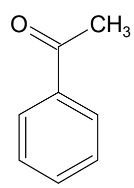
(B)  (Both X and Y)

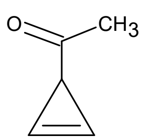
(C)  and 

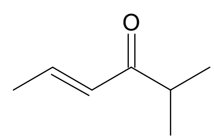
(D) None of the above

22. Which of the following will be most acidic?

(A) 

(B) 

(C) 

(D) 

Space for rough work

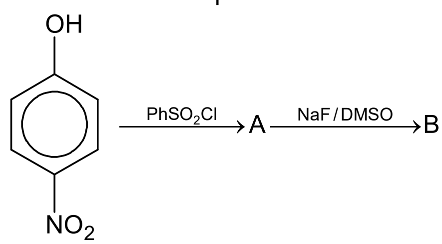
23. Equal weights of methane and hydrogen are mixed in an empty container at 25°C. The fraction of total pressure exerted by hydrogen is

(A) $\frac{1}{2}$
(B) $\frac{8}{9}$
(C) $\frac{1}{9}$
(D) $\frac{16}{17}$

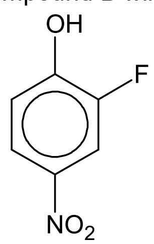
24. The ratio of rate of diffusion of H_2 and He is 1:2, then the composition of the gas mixture initially effusing out will be in the ratio of

(A) $1:2\sqrt{2}$
(B) $\sqrt{2}:1$
(C) 2:1
(D) 1:2

25. For the reaction sequence:



The compound B will be

(A)


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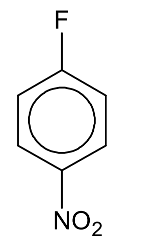
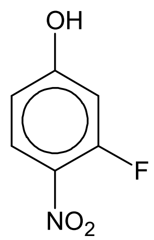


(C)


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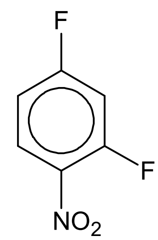
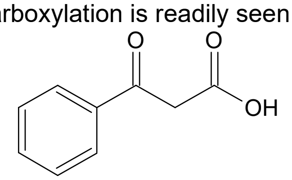
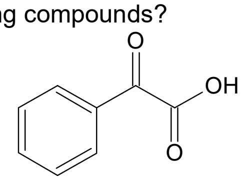
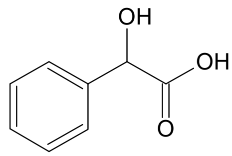


Space for rough work

26. Decarboxylation is readily seen in which of the following compounds?

(A)


(B)


(C)


(D) Both (A) and (B)

27. The oxidation number of sulphur in S_8, S_2F_2, H_2S respectively, are
(A) 0, +1 and –2
(B) +2, +1 and –2
(C) 0, +1 and +2
(D) –2, +1 and –2

28. How many grams of $K_2Cr_2O_7$ is present in 1 litre of its N/10 solution in acid medium?
(A) 4.9
(B) 49
(C) 0.49
(D) 3.9

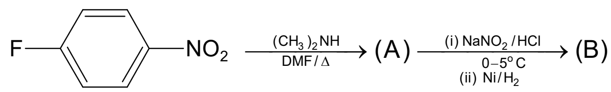
29. The % of free SO_3 in oleum that is labelled 109% H_2SO_4 by weight is
(A) 40
(B) 60
(C) 20
(D) 70

30. The intensive property is
(A) ΔU
(B) ΔH
(C) ΔG
(D) $C_{p,m}$

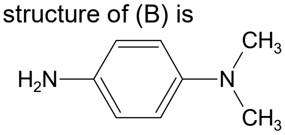
31. Heat of neutralization of four monobasic acids A, B, C and D are –13.7, –9.4, –11.2 and –12.4 kcal/mole respectively, when they are neutralized by a common base. The acidic character obeys the order,
(A) A > B > C > D
(B) A > D > C > B
(C) D > C > B > A
(D) D > B > C > A

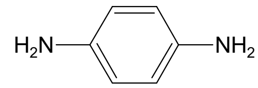
32. Bond energy of N ≡ N, H – H bond, N – H bond are a, b and c respectively. The ΔH for the reaction, $2NH_3 \longrightarrow N_2 + 3H_2$ is
(A) 6c – 3b – a
(B) 6c + 3b + a
(C) c + 6b – a
(D) 6c + b – 3a

Space for rough work

33.


The structure of (B) is

(A)


(B)


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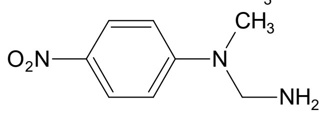
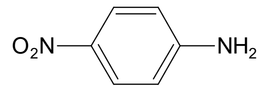
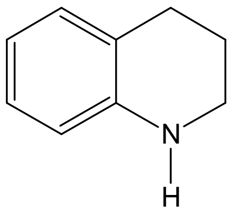


(D)


34. Which is maximum basic in nature?

(A)


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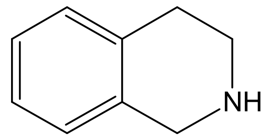
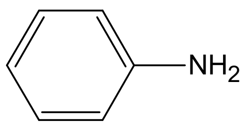


(C)


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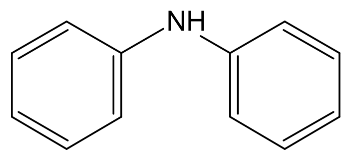
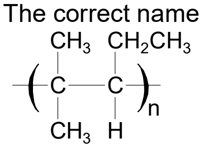


35. Osazone formation is used to characterize

(A) polymers
(B) sugars
(C) carboxylic acid
(D) alcohols

36. The correct name for the alkene monomer which forms the polymer shown below, is



(A) 2-methyl-3-ethylpropene
(B) 2-methylpent-2-ene
(C) 2-methylpent-3-ene
(D) 4-methylpent-2-ene

37. Cyanide process is applicable for the extraction of

(A) Ag
(B) Ca
(C) Fe
(D) Mg

Space for rough work

38. Which of the following acid contains one S – S bond?
(A) $H_2S_2O_7$
(B) $H_2S_2O_8$
(C) $H_2S_2O_6$
(D) none of these

39. Structure of $Na_2B_4O_7.10H_2O$ contains
(A) Two triangular and two tetrahedral units
(B) Three triangular and one tetrahedral units
(C) All tetrahedral units
(D) All triangular units.

40. In P_4O_{10} the number of P – O – P bonds are
(A) 4
(B) 6
(C) 8
(D) 10

41. The complex $[Cr(H_2O)_4Br_2]Cl$ give test for
(A) Br^-
(B) Cl^-
(C) Cr^{+3}
(D) Cr^{+3}, Br^- and Cl^-

42. If co-ordination number of $[M^{2+}]$ is four, how many moles of acetylacetonate anion is needed to form stable complex
(A) one
(B) two
(C) three
(D) four

43. The half-life period of the decomposition of a compound is 20 minutes. If the initial concentration is doubled, the half-life period is reduced to 10 mins. What is the order of the reaction?
(A) 1
(B) 2
(C) 3
(D) 4

44. The Molecularity of a reaction is:
(A) Always two
(B) Same as its order
(C) Different than the order
(D) May be same or different as compared to order

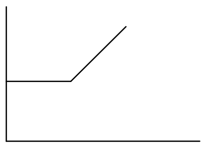
45. A decinormal solution of potassium ferrocyanide is 25% dissociated at 300^0K. Calculate osmotic pressure of the solution.
(A) $1.247 \times 10^5 Nm^{-2}$
(B) 1.247×10^2 atm
(C) 1.247×10^3 mm of Hg
(D) 1.247×10^3 atm

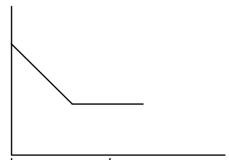
Space for rough work

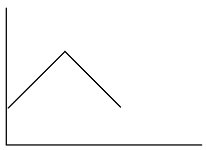
46. Given the cell: Cd(s) |$Cd(OH)_2$(s)|NaOH(aq, 0.01 M) | H_2(g, 1 bar| Pt(s) with E_{cell} = 0.0 V. If $E^0_{Cd^{2+}|Cd}$ = –0.39V, then K_{sp} of $Cd(OH)_2$ is :

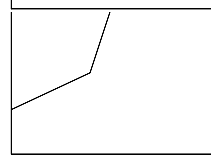
(A) 0.1
(B) 10^{-13}
(C) 10^{-15}
(D) none of these

47. CH_3COOH is titrated with NaOH solution and the conductance of CH_3COOH solution is measured each time after addition of certain amount of alkali. The titration curve obtained by plotting conductance along Y–axis and volume of NaOH solution added along X–axis, will be like.

(A) 

(B) 

(C) 

(D) 

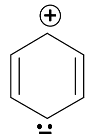
48. What is the number of atoms in a cubic based unit cell having one atom on each corner and two atoms on each body diagonal?

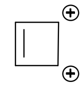
(A) 8
(B) 9
(C) 7
(D) 10

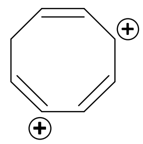
Space for rough work

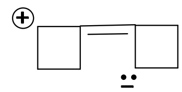
ANY NUMBER OF OPTIONS 4, 3, 2 or 1 MAY BE CORRECT
MARKS WILL BE AWARDED ONLY IF ALL THE CORRECT OPTIONS ARE BUBBLED

49. Which of the following species is/are aromatic?

(A) 

(B) 

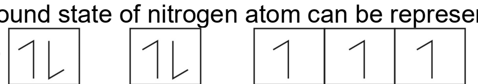
(C) 

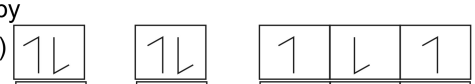
(D) 

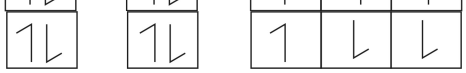
50. The time required for an electron to make one complete revolution around nucleus of H-atom in a higher orbit n_2 is 8 times to that of a lower orbit n_1. Therefore n_1 and n_2 are

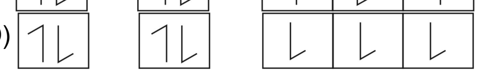
(A) 1 and 2
(B) 2 and 3
(C) 2 and 4
(D) 3 and 6

51. Ground state of nitrogen atom can be represented by

(A) 

(B) 

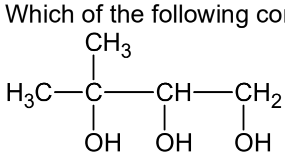
(C) 

(D) 

52. IE_2 for an element are invariably higher than IE_1

(A) The size of cation is smaller than its atom
(B) It is difficult to remove 'e' from cation
(C) IE is endothermic
(D) All of the above

53. Which of the following compounds are formed by the oxidation of



with HIO_4

(A) CH_3-CHO
(B) CH_3COCH_3
(C) $H-COOH$
(D) HCHO

54. In the following reaction

$$\underset{(A)}{\left[Cu(H_2O)_3(OH)\right]^{+}}+\underset{(B)}{\left[Al(H_2O)_6\right]^{3+}}\rightarrow\underset{(C)}{\left[Cu(H_2O)_4\right]^{2+}}+\underset{(D)}{\left[Al(H_2O)_5(OH)\right]^{2+}}$$

(A) A is an acid and B the base
(B) A is a base and B the acid
(C) C is the conjugate acid of A, and D is the conjugate base of B
(D) C is conjugate base of A, and D is the conjugate acid of B

Space for rough work

55. Which of the following is correct regarding strength of hydrogen bond?
(A) $F-H\cdots\cdots F < O-H\cdots\cdots O$
(B) $F-H\cdots\cdots F > O-H\cdots\cdots O$
(C) $F-H\cdots\cdots F > Cl-H\cdots\cdots Cl$
(D) $O-H\cdots\cdots O > N-H\cdots\cdots N$

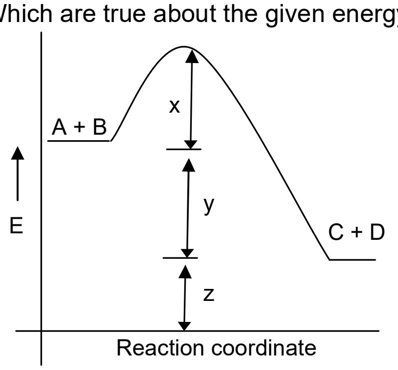
56. Which combination (s) given below is / are correct?
(A) $HgCl_2$ – Linear
(B) ClF_3 – T shaped
(C) ICl_4^- - square planar
(D) XeF_6 – pentagonal bypyramidal

57. Which is/are correct?
(A) $2.303\log K = -\frac{\Delta H^o}{RT} + \frac{\Delta S^o}{R}$
(B) $\Delta G^o = -2.303RT\log K$
(C) $-2.303\log K = -\frac{\Delta H}{RT^2} + \frac{\Delta S^o}{R}$
(D) $2.303\log K = \frac{1}{RT}\left(\Delta H^o + \Delta S^o\right)$

58. Which of the following is/are correct regarding graphite:
(A) good conductor of electricity
(B) delocalised π electrons are present
(C) allotrope of carbon
(D) poor conductor of electricity

59. Which of the following compounds will show geometrical isomerism?
(A) $[Pt(NH_3)_2Cl_2]$
(B) $[Co(NH_3)_5(NO_2)]Cl_2$
(C) $[Co(en)_2Cl_2]Cl$
(D) $[Co(NH_3)_4Cl_2]Cl$

60. Which are true about the given energy diagram:



(A) The enthalpy change of the reaction A + B → C + D is represented by the value “y”
(B) The activation energy of the reaction A + B → C + D is “x + y”
(C) The activation energy of the reaction C + D → A + B is “x + y + z”
(D) In presence of the positive catalyst the activation energy for the conversion A + B → C + D is somewhat less than x

Space for rough work

ANSWERS

1.	**C**	2.	**A**	3.	**D**	4.	**B**
5.	**B**	6.	**A**	7.	**D**	8.	**D**
9.	**B**	10.	**A**	11.	**C**	12.	**A**
13.	**A**	14.	**C**	15.	**B**	16.	**A**
17.	**B**	18.	**B**	19.	**B**	20.	**D**
21.	**C**	22.	**A**	23.	**B**	24.	**A**
25.	**B**	26.	**A**	27.	**A**	28.	**A**
29.	**A**	30.	**D**	31.	**B**	32.	**A**
33.	**A**	34.	**B**	35.	**B**	36.	**B**
37.	**A**	38.	**C**	39.	**A**	40.	**B**
41.	**B**	42.	**B**	43.	**B**	44.	**D**
45.	**A**	46.	**C**	47.	**D**	48.	**B**
49.	**A, B**	50.	**A, C, D**	51.	**A, D**	52.	**A, B**
53.	**B, C, D**	54.	**B, C**	55.	**B, C, D**	56.	**A, B, C**
57.	**A, B**	58.	**A, B, C**	59.	**A, C, D**	60.	**A, D**

HINTS & SOLUTIONS

5. $K.E. = h\nu - h\nu_o = \frac{hc}{\lambda} - h\nu_o$ and then $\ell = \frac{h}{\sqrt{2K.E.(M)}}$

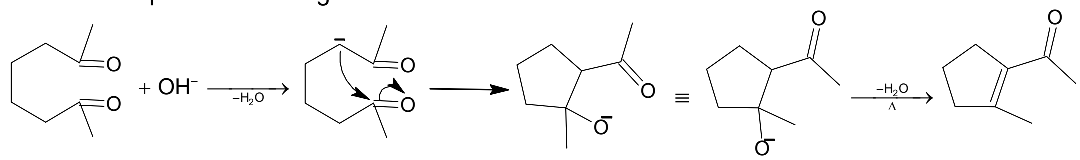
11. Cyclic epoxides are used to add – C – atoms at the same time at an alkyl group through Grignard's Reagent.

13. $K_p = \frac{[SO_3]^2}{[O_2][SO_2]^2} = 5$

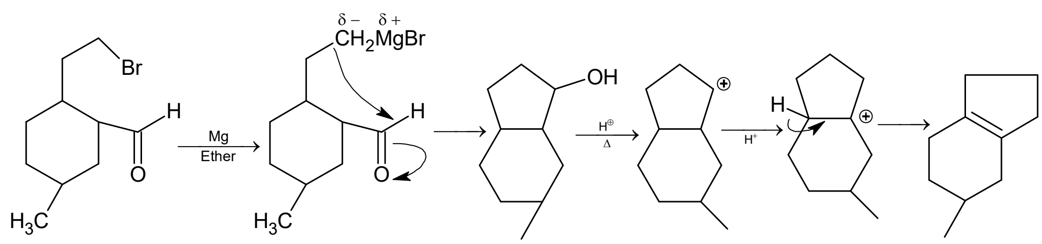
$\Rightarrow [O_2] = 0.2$

14. Kw = $10^{-8} \times 10^{-8}$
= 10^{-16}
$[H^+][OH^-] = 10^{-16}$
$[H^+] = 10^{-16}/10^{-6} = 10^{-10}$

19. The reaction proceeds through formation of carbanion.

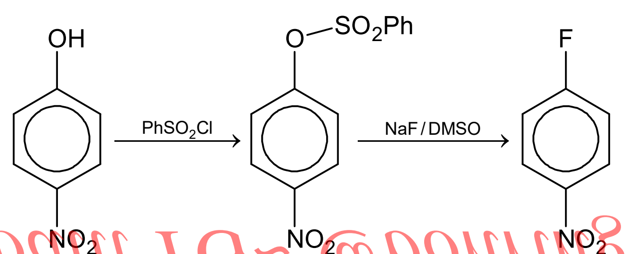


20.



22. Ion becomes aromatic.

25.



(A) (B)

F^- is very reactive unsolvated which can displace $PhSCO_3^-$ which being a good leaving group in presence of strong withdrawing group —NO_2.

28. Number of equivalent of $K_2Cr_2O_7 = \frac{1}{10} \times 1$

weight = $\frac{1}{10} \times \frac{M}{6}$

31. In case of weak acid, heat of neutralization is greater than –13.7 kcal/mole because extra energy is required to dissociate the acid. Hence, (B).

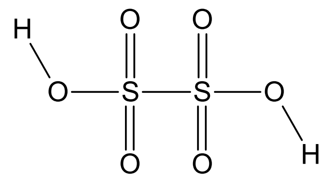
32. ΔH reaction = $\sum$BE reactant – $\sum$BE product

$= 2\times 3c - a - 3b$

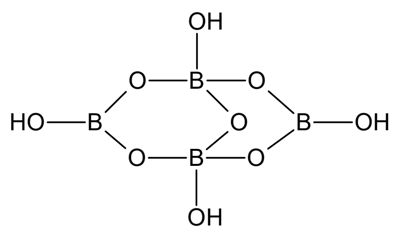
$= 6c - 3b - a$

Hence, (A).

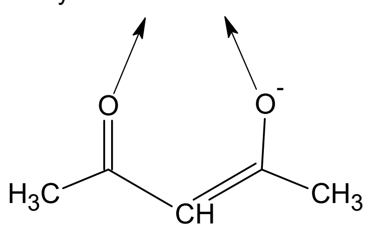
38.



39.



42. Acetyl acetate anion is bidentate ligand, so two moles are needed



Hence (B)

43. $t_{1/2}$ = 20 min; $t'_{1/2} = 10\,\text{min}$ $[A]'_o = 2\times[A]_o$

$\because t_{1/2} \propto \dfrac{1}{[A]_o^{n-1}}$

$$\frac{t_{1/2}}{t'_{1/2}} = \left[\frac{[A]'_o}{[A]_o}\right]^{n-1}$$

$$\frac{20}{10} = \left[\frac{2\times[A]_o}{[A]_o}\right]^{n-1}$$

$(2)^1 = 2^{n-1}$

$n - 1 = 1$

$n = 1 + 1 = 2$

45. For $K_4[(Fe(CN)_6)]$ n factor is 4.

Molarity $= \dfrac{1}{10}\times\dfrac{1}{4} = \dfrac{1}{40}\times 10^3\,\text{mol}/\text{m}^3$

$\therefore \pi v = \text{nst}/p = 6.236\times 10^4\,\text{Mn}^{-2}$.

$\therefore \alpha = 25\% = .25$

$i = \dfrac{\pi_{exp}}{\pi_N} = (1+4\alpha)$

$p_{exp} = 6.236\times 10^4(1+4\times 0.25)$

$= 1.247\times 10^5\,\text{N}/\text{m}^2$

46. $$E_{cell} = E_{H^+/H_2} - E_{OH^-/cd(OH)_2/cd}$$

$$= E_{H^+/H_2} - E_{cd^{2+}/cd}$$

$$E_{cell} = 0$$

$$E^0_{cell} = 0.03^2 \log \frac{[cd^{2+}][OH^-]}{K\omega^2}$$

$$\log \frac{k_{SP}}{K\omega^2} = \frac{0.39}{0.03} = 13$$

$$K_{SP} = 10^{-15}$$

48. There are 4 diagonals in a cube. As there are two atoms on each diagonals, here a total of 4 × 2 = 8 atoms on the diagonal.

Contribution of 8 corner atoms = $8 \times \frac{1}{2}$ = 1 atom.

∴ Total no. of atoms in the unit cell = 8 + 1 = 9 atoms.

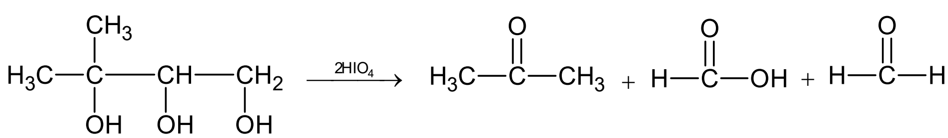
50. Time period ∝ n^3

$$\frac{T_1}{T_2} = \left(\frac{n_1}{n_2}\right)^3$$

$T_2 = 8\ T_1 \Rightarrow n_2 = 2n_1$

51. Hund's rule.

53.



58. Graphite is good conductor of electricity, due to delocalised π electrons, and it is allotrope of carbon.

60. From the given diagram ΔH for the reaction A + B → C + D is "y". The activation energy for A + B → C + D is 'x', and it is less than 'x' in presence of +ve catalyst.